

Sympathetic nervous system activity and resting metabolic rate in vegetarians.

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We examined the resting metabolic rate (RMR) and sympathetic nervous system activity of young male vegetarians (n = 17) and nonvegetarians (n = 40). Subjects were characterized for RMR by indirect calorimetry, norepinephrine kinetics from infusions of tritiated norepinephrine, energy and macronutrient intake from a 3-day food diary, and body composition by underwater weighing. Vegetarians reported a greater relative intake of carbohydrates (62% $\pm$ 5% v 51% $\pm$ 6%, $P < .01$) and a lower relative intake of fat (25% $\pm$ 5% v 33% $\pm$ 6%, $P < .01$) than nonvegetarians, whereas no differences were observed in daily energy intake, body composition, or maximal aerobic capacity (VO_{2max}) between groups. Vegetarians exhibited an 11% higher absolute RMR (1.29 $\pm$ 0.15 v 1.16 $\pm$ 0.13 kcal/min, $P < .01$), a higher plasma concentration of norepinephrine (216 $\pm$ 33 v 165 $\pm$ 18 pg/mL, $P < .01$), and a greater norepinephrine appearance rate (0.50 $\pm$ 0.08 v 0.36 $\pm$ 0.09 micrograms/min, $P < .01$) than nonvegetarians. After statistically controlling for differences in relative amounts of carbohydrate and fat in the diet and for norepinephrine concentrations, no significant differences in adjusted RMR between vegetarians and nonvegetarians were noted. These results suggest that the higher RMR observed in young male vegetarians is partially mediated by differences in dietary macronutrient composition and increased sympathetic nervous system activity.